

**ASSESSMENT SPECIFICATION
PYTHON IN FINANCE, FINANCE IN PYTHON
JAGIELLONIAN UNIVERSITY**

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OVERVIEW

The aim of the assessment is to design and implement a quantitative finance framework in Python combining derivative pricing, interest-rate modelling, calibration, and numerical methods.

The project should demonstrate both understanding of the underlying financial mathematics and the ability to produce clean, maintainable Python code.

The submission should be prepared as a Git repository and shared with the lecturer via GitHub, GitLab, or Bitbucket as either a private or public repository.

After submission, each student (or group) will participate in a short Microsoft Teams discussion explaining the modelling and implementation choices.

CORE REQUIREMENTS

The project must contain:

1. A CRR-type tree pricing framework.
2. At least two interest-rate models.
3. Calibration of the interest-rate models to market data that will be provided later or appropriate data you have found yourself.
4. Numerical experiments and comparison studies.
5. Documentation and example usage.

The implementation should be modular and extensible rather than consisting of a single monolithic notebook.

Students are encouraged to use:

- `numpy`,
- `scipy`,
- `pandas`,
- `matplotlib`,
- Jupyter notebooks,
- `pytest`.

CRR TREE MODELS

Students must implement CRR tree models capable of pricing at least:

- European options,
- American options,
- Bermudan options.

The framework should be sufficiently flexible to support exotic instruments. A payoff specification will be provided later during the course, so the architecture should allow additional payoffs to be added without rewriting the pricing engine.

There should also be meaningful flexibility in the construction of the trees. Examples include different choices of:

- up/down factors,
- probability constructions,

- discretisation schemes,
- convergence approaches.

Students are encouraged to compare the numerical behaviour of different modelling choices.

INTEREST-RATE MODELS

Students must implement at least two interest-rate models. Examples include:

- Vasicek,
- CIR,
- Hull–White,
- Ho–Lee,
- Black–Karasinski,
- BDT.

Other models may also be used if appropriately justified.

The models should be calibrated to market data that will be provided later. The project should discuss:

- calibration targets,
- optimisation methods,
- numerical stability,
- interpretation of fitted parameters.

At least one interest-rate model should include comparison of different discretisation or simulation approaches.

REPOSITORY EXPECTATIONS

The repository should contain:

- organised source code,
- tests,
- example notebooks,
- documentation,
- clear instructions for running the project.

A typical structure could look as follows:

```
project/
```

```
src/
tests/
notebooks/
data/
figures/
README.md
requirements.txt
```

however it is not a strict requirement. Code should contain reasonable documentation, comments, and type annotations where appropriate.

NUMERICAL EXPERIMENTS

The project should contain a non-trivial numerical section. This may include:

- convergence studies,
- calibration diagnostics,
- sensitivity analysis,
- comparison of modelling assumptions,
- runtime analysis,
- comparison against analytical formulas where available.

Good plots and visual presentation are strongly encouraged.

TEAMS DISCUSSION

After submission there will be a short Teams discussion concerning:

- modelling assumptions,
- implementation decisions,
- calibration methodology,
- numerical choices,
- strengths and limitations of the framework.

Students should be able to explain and defend their design choices.

ASSESSMENT CRITERIA

Indicative evaluation criteria are:

Criterion	Weight
Correctness of financial models	25%
Code quality and repository structure	20%
Calibration methodology	15%
Numerical experiments and analysis	15%
Flexibility and extensibility	10%
Teams discussion	10%
Documentation and presentation	5%

ADDITIONAL REMARKS

There will be no Value-at-Risk or Expected Shortfall calculations in this year's project in order to keep the scope manageable and allow greater focus on pricing models and interest-rate modelling.

Strong projects will typically include:

- careful software architecture,
- reusable pricing components,
- meaningful model comparisons,
- well-designed numerical experiments,
- clear financial interpretation.

As a general illustration of the expected ambition and repository quality, students may look at projects such as

<https://github.com/OneAndZero24/irm>

although the present year's project should not simply replicate that repository.

Students may use publicly available references and documentation, but all submitted work must reflect genuine understanding.

The use of AI tools is permitted only as an auxiliary aid. Students remain fully responsible for the correctness of the implementation and for explaining the mathematical and numerical aspects of their project during the Teams discussion.

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